



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

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Protocol / Portal/Application/URL

❖ URL (Uniform Resource Locator)

- A **Uniform Resource Locator (URL)** is a way of identifying the location of a file on the internet. They're what we use to open not only websites, but also to download images, videos, software programs, and other types of files that are hosted on a server.
- **URL (Uniform Resource Locator)** is the unique address of a resource on the Internet/Web, such as a webpage, image, video, or file. It tells the browser where and how to access a particular resource.
- It is also known as an **internet address** or **web address**.
- A website contains many webpages/resources, and each resource can have its own URL.
- Example of URL : <https://computer24y7.github.io/home/part2.html>

❖ How to identify website and URL

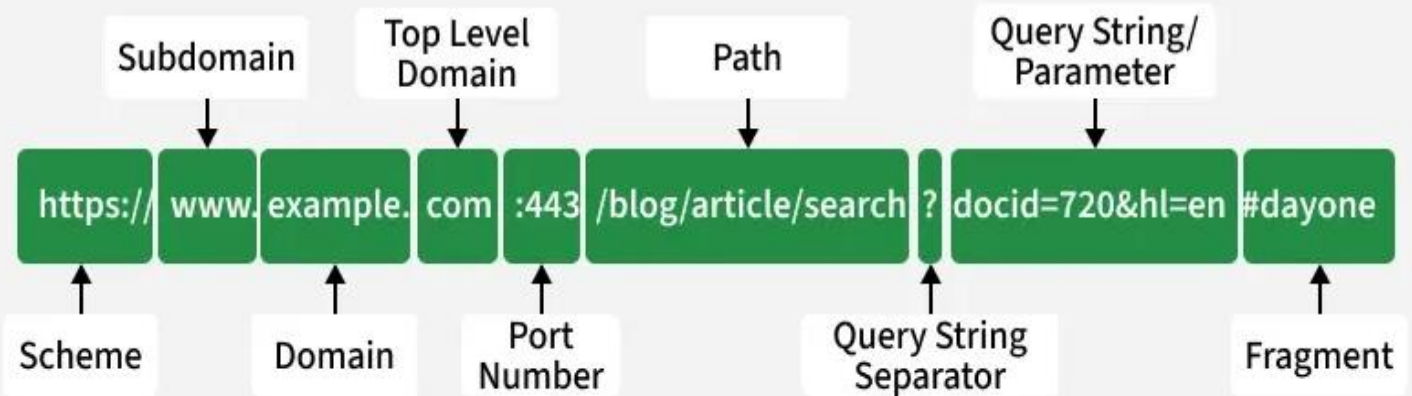
	Entire Website	URL
www.example.com	Yes	Yes
www.example.com/contact.html	No	Yes
https://computer24y7.github.io/home/part2.html	No	Yes
Example.com	Yes	yes

- Above table, Every website has one or more URLs, but not every URL is an entire website.

❖ Component of URL

URL Parts

URL: <https://www.example.com:443/blog/article/search?docid=720&hl=en#dayone>



i. Protocol / Scheme

- This indicates, which protocol how are data transfer b/w browser and server/resource.
- E.g. – http , https, ftp,.....

ii. Subdomain

- Identify a distinct/different section or service within the domain.
- Example : <https://blog.example.com>
Here, blog → subdomain, example → main domain, .com→Tld
- **Note** : one domain can be different type of subdomain.
- Example :
- www.example.com, blog.example.com, shop.example.com, mail.example.com

iii. Domain name

- This is a name and address of website.
- E.g – www.google.com → DNS → 8.8.8.8 or 8.8.4.4
- <https://computer24y7.github.io>

iv. Port name

- It identifies a specific network service/application on the server.
- E.g – http (80 port) , Https (443 port), dns (50 port)

v. Path

- It is identify address of specific file/folder/page inside of webserver (just like your computer search file/folder using path)
- E.g - <https://admission.ppuonline.in/Login.aspx> (in web)
C:\Program Files\CodeBlocks (in your pc)

vi. Query String / Query parameters

- Give extra/additional information of web page.
- E.g. –
https://serviceonline.bihar.gov.in/renderApplicationForm.do?serviceld=4630012&UUID=8f0c0826-8064-4687-a636-baa78a0cd505&directService=true&templd=751&grievDefined=0&serviceLinkRequired=No&userLoggedIn=N&source=CTZN&OWASP_CSRFTOKEN=AYGG-5P75-6NUX-TKNW-5713-JO8U-FZ7B-D51X
 - ✓ ? = starting point of query string
 - ✓ & = divided of two parameters
- A **query parameter** generally has this form: **parameter_name=value**
 - ✓ serviceld=4630012 → first query parameter
 - ✓ Uuid=8f0c0826..... → second query parameter,....total 9 parameter

Note : Website developer अपनी जरूरत के अनुसार parameters बना सकता है—जैसे id, page, search, category, user, token आदि।

vii. Fragment

- Indicates specific section/anchor of within/same web page. (लंबे webpage के किसी खास section में जाने के लिए)
- It is start →"#"
- E.g. - #section1, #comments,.....

❖ Inside of URL

- protocol + subdomain+domain+TLD+port+ path+ query parameters + Fragment
- Note : It is not necessary for every URL to contain all these components.

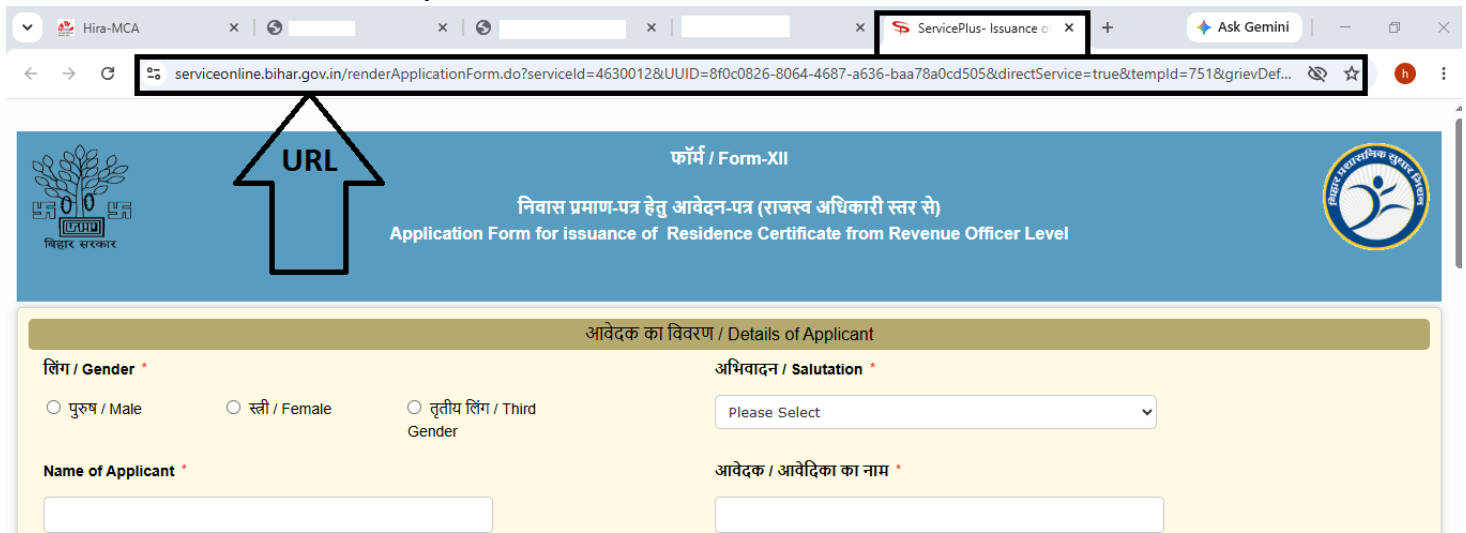
But 2(two) component are essential/mandatory

- (i) scheme(protocol)
- (ii) host/domain.

Example : [https:// example.com](https://example.com)

❖ Where is the URL located?

A URL is entered/displayed in the address bar of a web browser. You can see this URL at the top of the browser window, in the address bar.



❖ Application

- An application (app) is a software program designed to perform a specific task for the user.
- Application run on device like computer, smartphone, tablets. Mobile,.... and can be installed or accessed online/offline.
- Example : Whatsapp, chrome, ms-word, calculator, youtube,....

Types of Application

Based on internet

- (i) Offline application
Without internet use.
Example : Calculator, notepad, photoshop, ms-word,.....
- (ii) Online application
Use internet
Example : Gmail, online banking, phonepay, whatsapp.....
- (iii) Hybrid application
Use both offline and online
Example : Microsoft 365, google map, Google Keep, Spotify /

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YouTube Music,.... (data should be downloaded in app)

Based on Platform (hardware)

- (i) Desktop application : Windows, macOS/ Linux,...
Example of app : MS word, VLC,.....
- (ii) Mobile application : Android, IOS,...
Example : Whatsapp, Instagram,....
- (iii) Web application :
 - A web application is an **interactive software** application that runs on a web server and is accessed through a web browser over a network or the Internet.
 - simple website that mainly provides information, a web application allows users to **interact with the application and perform tasks**.
 - Example : amazon app (application) & amazon.com(website),
Gmail (application) & gmail.com (website),....

Based on purpose

- (i) Educational application : e.g. unacademy, adda, pw, duolingo...
- (ii) Business application : e.g. Tally, zoho books,...
- (iii) Financial application : e.g. Banking apps, gpay, phonepay,...
- (iv) Productivity application : e.g. Ms word, excel, powerpoint,...
- (v) Communication application : e.g. whatsapp, gmail, zoom,...
- (vi) E-commerce application : e.g. amazon, flipkart,....
- (vii) Development/programming app : VS code, codeblock,...
- (viii) Security application : Antivirus, password managers,...
- (ix) Navigation/travel application : Google maps, makemytrip,...
- (x) Healthcare application : telemedicine,....
- (xi) Scientific/engineering app : AutoCAD, MATLAB,...
- (xii) Multimedia application : VLC, Video editors,.....
- (xiii) Entertainment application : Youtube, spotify, Netflix,...
- (xiv) Social media application : Instagram, facebook,...
- (xv) Graphics/design application : Photoshop, canva,....

❖ Portal

- A portal is a special type of website or web application that serves as a single access point to a variety (many) of information, services, or

resources.

- It usually requires login and provides personalized content based on the user's profile
- Centralized access to multiple services
- Often given User authentication (login)
- Different sections like news, forms, applications, results, etc.

Example :-

1. india.gov.in (government service portal / National Portal of India)
2. University student portal – for examdate, Examforms, admission, result,....



Note : All portal is called website, but all website is not called portal.

❖ Protocols

- A protocol is a set of rules that governs communication between devices/computers over a network.
- These rules define how data is formatted, transmitted, received, and processed between communicating devices.

❖ Main purposes of protocols

1. **Data formatting** – Defines how data should be structured/formatted before transmission.

(Data को किस format में भेजना है।)

2. **Data transmission** – Defines how data is sent from one device to another.

(Data को एक device से दूसरे device तक कैसे भेजना है।)

3. **Addressing** – Identifies the sender and receiver of data.

(Sender और receiver की पहचान करना।)

4. Error detection and handling – Helps detect and handle errors during communication.

(Data transmission में errors को detect और handle करना।)

5. Flow control – Controls the rate of data transmission so that the receiver is not overwhelmed.

(Data भेजने की speed/rate को control करना।)

6. Acknowledgement – Allows the receiver to indicate that data has been received.

(Receiver द्वारा data receive होने की confirmation देना।)

7. Synchronization / sequencing – Helps maintain the correct order and timing of data.

(Data को सही sequence और timing में maintain करना।)

8. Connection management – Defines how a connection is established, maintained, and terminated.

(Connection को establish, maintain और terminate करना।)

❖ What is port number ?

- A port number is a 16-bit logical number used by TCP or UDP to identify a particular application/service on a device.
- It helps the operating system deliver incoming data to the correct application.
- Example : HTTP → TCP Port 80, HTTPS → TCP Port 443, DNS → UDP Port 53, SSH → TCP Port 22

❖ What is IP protocol number ?

- An IP Protocol Number is a number in the IP header that identifies the protocol carried inside an IP packet.
- It tells the receiving device what protocol should process the packet's payload.
- Example : ICMP → 1, TCP → 6 , UDP → 17,...

Short point

- ✓ IP Protocol Number → Which network/transport protocol?
- ✓ Port Number → Which application/service?
- ✓ IP address → Which device/server?

❖ Types of Protocol

No.	Protocol	Full Form	Port No.	IP Protocol No.	Work	काम
1	HTTP	HyperText Transfer Protocol	TCP 80	—	Transfers web resources	Web pages/resources transfer करना
2	HTTPS	HyperText Transfer Protocol Secure	TCP 443	—	Secure web communication	Secure web communication करना
3	FTP	File Transfer Protocol	TCP 20/21	—	Transfers files	Files transfer करना
4	SFTP	SSH File Transfer Protocol	TCP 22	—	Secure file transfer	Secure तरीके से files transfer करना
5	SSH	Secure Shell	TCP 22	—	Secure remote login	Secure remote login/command execution
6	Telnet	Teletype Network	TCP 23	—	Remote terminal access	Remote system पर terminal access
7	SMTP	Simple Mail Transfer Protocol	TCP 25, 587	—	Sends email	Email भेजना
8	DNS	Domain Name System	UDP/TCP 53	—	Resolves domain names to IP addresses	Domain name को IP address में resolve करना
9	DHCP	Dynamic Host Configuration Protocol	UDP 67/68	—	Automatically assigns IP configuration	Automatically IP configuration देना

10	POP3	Post Office Protocol Version 3	TCP 110/995	—	Downloads email	Server से email download करना
11	IMAP	Internet Message Access Protocol	TCP 143/993	—	Accesses/manages email on server	Server पर email access/manage करना
12	SNMP	Simple Network Management Protocol	UDP 161/162	—	Network monitoring/management	Network devices monitor/manage करना
13	NTP	Network Time Protocol	UDP 123	—	Synchronizes time	Network devices का time synchronize करना
14	LDAP	Lightweight Directory Access Protocol	TCP/UDP 389	—	Accesses directory services	Directory information access करना
15	LDAPS	LDAP over TLS	TCP 636	—	Secure directory access	Secure directory access
16	BGP	Border Gateway Protocol	TCP 179	—	Exchanges routing information between ASes	अलग-अलग networks/AS के बीच routing information exchange
17	RIP	Routing Information Protocol	UDP 520	—	Exchanges routing information	Routing information exchange करना
18	OSPF	Open Shortest Path First	—	89	Finds shortest/best routes	Best/shortest route calculate करना
19	EIGRP	Enhanced Interior Gateway Routing Protocol	—	88	Interior routing	Internal network routing करना

20	ICMP	Internet Control Message Protocol	—	1	Error/control/diagnostic messages	Error, control और diagnostic messages
21	IGMP	Internet Group Management Protocol	—	2	Manages IPv4 multicast membership	IPv4 multicast groups manage करना
22	TCP	Transmission Control Protocol	Uses application ports	6	Reliable connection-oriented delivery	Reliable, connection-oriented data delivery
23	UDP	User Datagram Protocol	Uses application ports	17	Connectionless transport	Connectionless data delivery
24	IPv4	Internet Protocol Version 4	—	4	Logical addressing and routing	Logical addressing और routing
25	IPv6	Internet Protocol Version 6	—	41	128-bit addressing and routing	128-bit addressing और routing
26	ARP	Address Resolution Protocol	—	—*	Finds MAC address from IPv4 address	IPv4 address से MAC address पता करना
27	RARP	Reverse Address Resolution Protocol	—	—*	Finds IP from MAC address	MAC address से IP पता करना
28	SIP	Session Initiation Protocol	UDP/TCP 5060; TLS 5061	—	Establishes/manages VoIP sessions	VoIP call/session establish/manage करना
29	RTP	Real-time Transport Protocol	Dynamic/commonly 1024–65535	—	Carries real-time audio/video	Real-time audio/video transport करना
30	RTCP	RTP Control Protocol	RTP-related ports	—	Monitors RTP sessions	RTP session की monitoring/control
31	RTSP	Real Time	TCP/UDP 554	—	Controls streaming media	Streaming media को control

		Streaming Protocol				करना
32	IRC	Internet Relay Chat	TCP 6667 (commonly)	—	Real-time text chat	Real-time text chat
33	NNTP	Network News Transfer Protocol	TCP 119/563	—	Transfers/distributes Usenet news	News/discussion messages transfer करना
34	TFTP	Trivial File Transfer Protocol	UDP 69	—	Simple file transfer	Simple file transfer
35	SMB	Server Message Block	TCP 445	—	File/printer sharing	File और printer sharing
36	NFS	Network File System	TCP/UDP 2049	—	Accesses remote filesystems	Remote filesystem access करना
37	MQTT	Message Queuing Telemetry Transport	TCP 1883/8883	—	IoT messaging	IoT devices के बीच messaging
38	CoAP	Constrained Application Protocol	UDP 5683/5684	—	Web-like communication for constrained IoT	Low-power IoT devices communication
39	RADIUS	Remote Authentication Dial-In User Service	UDP 1812/1813	—	Authentication/accounting	User authentication और accounting
40	TACACS+	Terminal Access Controller Access-Control System Plus	TCP 49	—	Network-device authentication/authorization	Network devices के लिए authentication/authorization
41	Kerberos	Kerberos Network	TCP/UDP 88	—	Network authentication	Network users/services को

		Authentication				authenticate करना
42	Syslog	System Logging Protocol	UDP 514; TCP 514/6514	—	Sends system logs	System logs भेजना
43	SCTP	Stream Control Transmission Protocol	No port itself	132	Reliable message-oriented transport	Reliable message-oriented transport
44	DCCP	Datagram Congestion Control Protocol	No port itself	33	Datagram transport with congestion control	Congestion control के साथ datagram transport
45	ESP	Encapsulating Security Payload	—	50	IPsec encryption/authentication	IPsec में encryption/authentication
46	AH	Authentication Header	—	51	IPsec authentication/integrity	IPsec authentication और integrity
47	GRE	Generic Routing Encapsulation	—	47	Encapsulates network protocols	Packets को encapsulate करना
48	L2TP	Layer 2 Tunneling Protocol	UDP 1701	—	Layer-2 tunneling	Layer-2 tunneling करना
49	PPTP	Point-to-Point Tunneling Protocol	TCP 1723 + GRE	47 (GRE)	VPN tunneling	VPN tunnel बनाना
50	ICMPv6	Internet Control Message Protocol Version 6	—	58	IPv6 error/control/diagnostic messages	IPv6 में error/control/diagnostic messages
51	RDP	Remote desktop	3389		Wireless control one device to another	
52	SNMP	Simple Network Management	161,162			Network device को manage और monitor करने के लिए ।

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